

AI CAN SEE THE PLASTIC. BUT CAN THE ROBOT PICK IT?

From AI detection to physical recovery of real post-consumer plastics,
at full industrial belt speed



Can accurate AI detections be converted into reliable physical recovery
on a moving industrial conveyor?

1 WHY THIS MATTERS



3.2 Mt

plastics reaching
end-of-life in Australia
(2023-24)

14%

national plastics
recovery rate

Recovering valuable material from complex
mixed waste remains difficult.

2 REAL ROBOT. REAL WASTE. REAL SPEED.

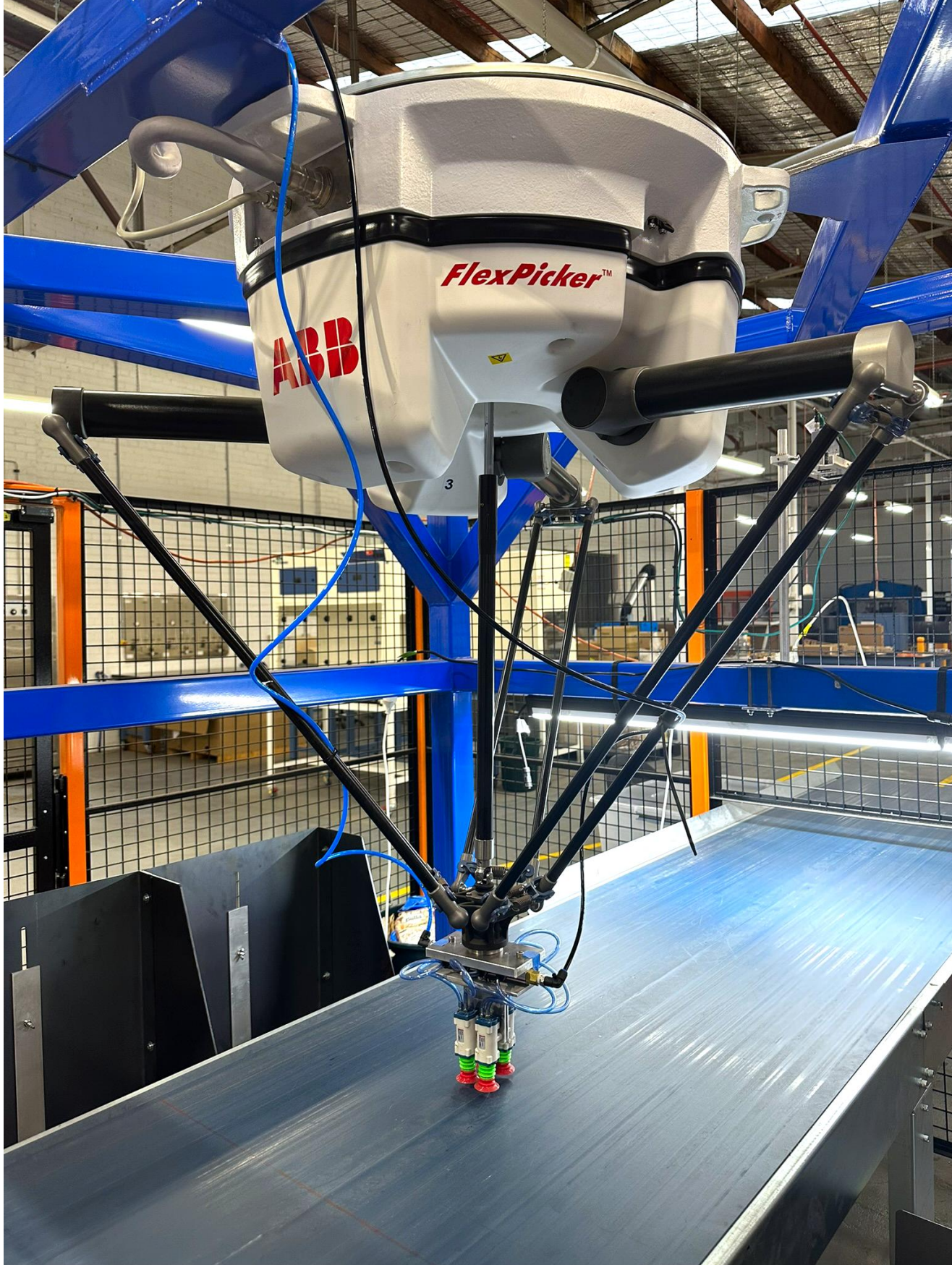
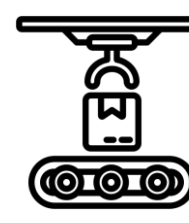


ABB FlexPicker IRB 360 delta robot with 4-cup suction gripper.
Tested at full industrial belt speed (0.88 m/s).

3 WHY THIS STUDY IS DIFFERENT

-  Real commercial post-consumer waste from an Australian MRF
-  Industrial ABB IRB 360 delta robot with 4-cup suction gripper
-  Validated at full industrial belt speed
-  End-to-end evaluation from AI vision to physical recovery

Previous peer-reviewed robotic waste-sorting studies operated at substantially lower conveyor speeds (up to 0.35 m/s).

VS

This study validated robotic sorting reaching an industrial belt speed of 1 m/s.

4 STUDY AT A GLANCE

-  20,100 images
-  40,628 annotated objects
-  11 waste classes
-  8 YOLO variants benchmarked
-  YOLOv11m selected and deployed
-  mAP@0.50 = 0.601 (standard detection performance score, 0-1)
-  5 classes x 5 cycles x 30 objects = 750 presentations
-  Robotic picking tests focused on five operationally important plastic categories

 First open Australian post-consumer plastic dataset, publicly released.

5 THE END-TO-END ROBOTIC SORTING PROCESS



6 MAIN FINDING

 4 of 5 deployment classes reached >96% precision (overall mAP@0.50 = 0.601 across all 11 classes)

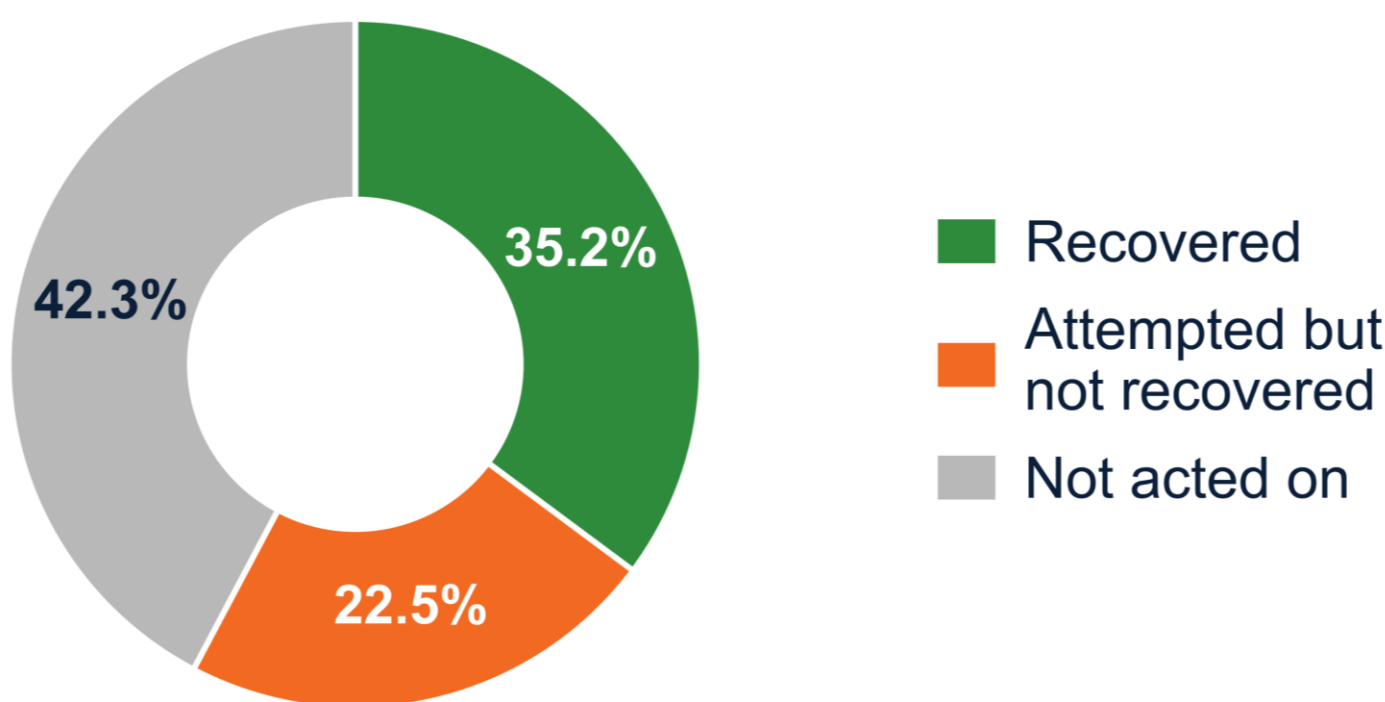


Per-class physical recovery ranged from 22.7% to 44.0%



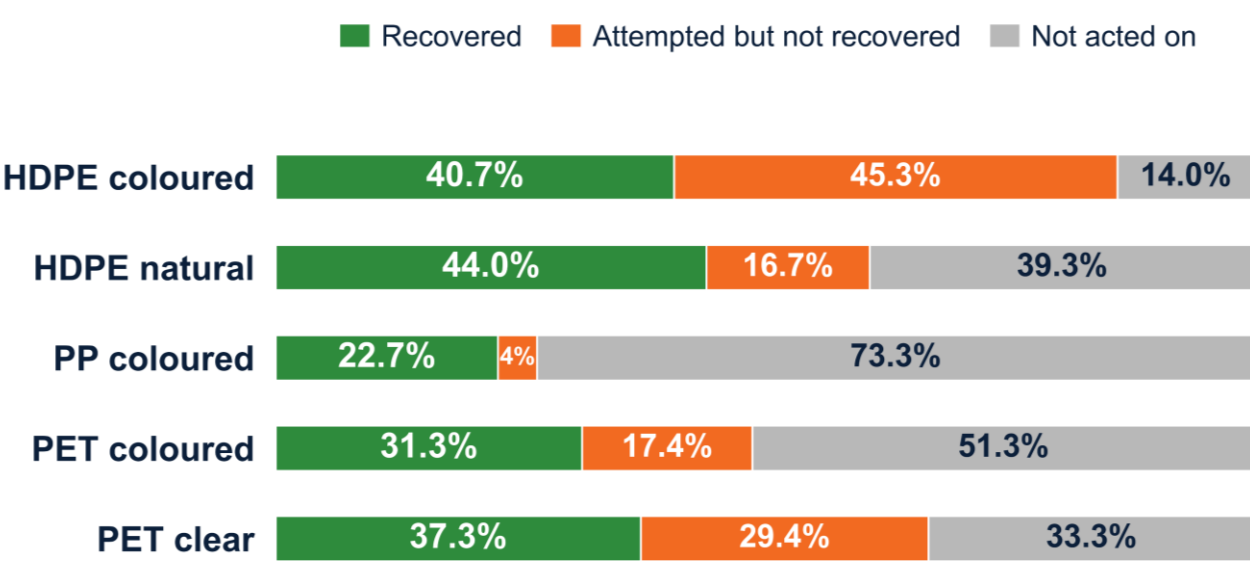
Accurate detection did not guarantee successful physical recovery.

7 WHERE WAS RECOVERY LOST OVERALL?



Results aggregated across all 5 target classes (750 object presentations).

8 WHERE DID OBJECTS GO?



Final recovery = attempt coverage x placement success.
Final recovery reflects both target coverage and successful placement once attempted.

9 WHY PERFORMANCE DROPPED

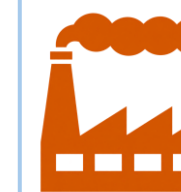
- GRIP**
 Irregular, crushed and dusty surfaces reduce suction reliability.
- TARGET AVAILABILITY & SCHEDULING**
 A single robot cannot act on every eligible target in time.
- VISION**
 RGB relies on visible appearance. Hidden labels and similar-looking plastics increase classification ambiguity.
RGB sees appearance, not polymer chemistry.

10 KEY TAKEAWAY DEPLOYMENT READINESS IS A CHAIN PROPERTY

High-performing AI is necessary, but reliable robotic sorting depends on seeing, tracking, selecting, scheduling, gripping and recovering the object as one coordinated system.



11 NEXT STEPS

-  **Adaptive gripping**
Improve grasp reliability on challenging items.
-  **Multi-object scheduling**
Increase target coverage with smarter planning.
-  **RGB-SWIR sensing**
Reduce material ambiguity and improve detection.
-  **Facility-scale validation**
Test long-term performance at industrial scale.

Hasala Pitiduwa Gamage

Department of Infrastructure Engineering,
The University of Melbourne
hasala.pitiduwa@student.unimelb.edu.au

Supervisors: Dr Shanaka Kristombu Baduge,
Prof. Priyan Mendis

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Scan for more details and project updates